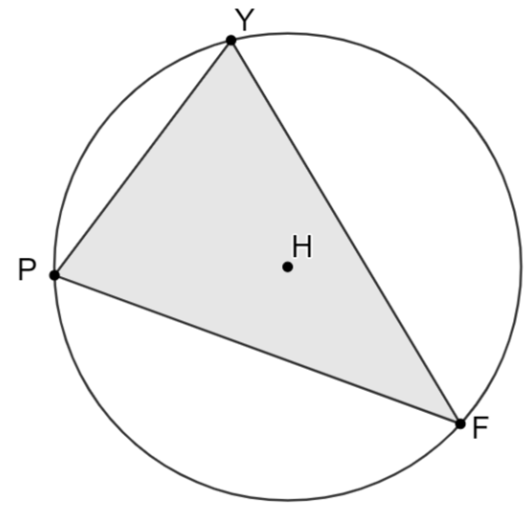
Geometry Honors
Unit 11
Calculator

Name____________________
Date________________
Period_________

1. Circle H is shown where points Y, F, and P are on the circle. $m\widehat{FY} = (11n + 14)°$, $m\angle PYF = (5n + 8)°$, and $m\angle YFP = (3n + 3)°$.



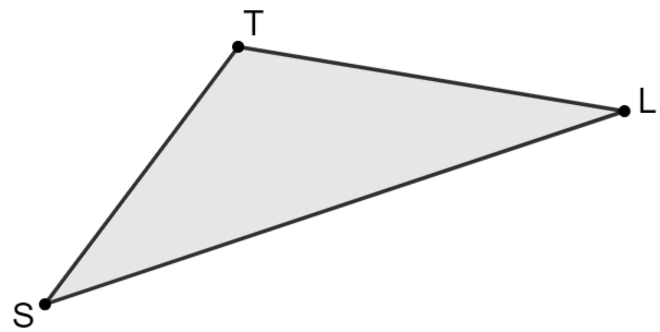
a. Determine the value of n.

$\boldsymbol{n =}$	

b. Determine $m\widehat{PY}$.

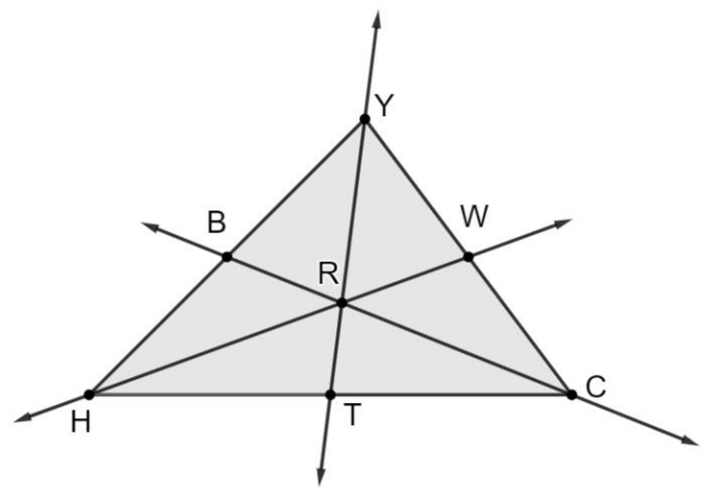
$\boldsymbol{m\widehat{PY} =}$	

2. Triangle TLS is shown where $TL = \frac{92}{15}$, $TS = 5$, and $SL = \frac{237}{25}$. List the angles in order from least to greatest.



Least		Greatest

3. Triangle YCH is shown where point R is the centroid. $RT = 13.4$, $RH = 39$, $YW = 25$, and $RB = 17.95$.



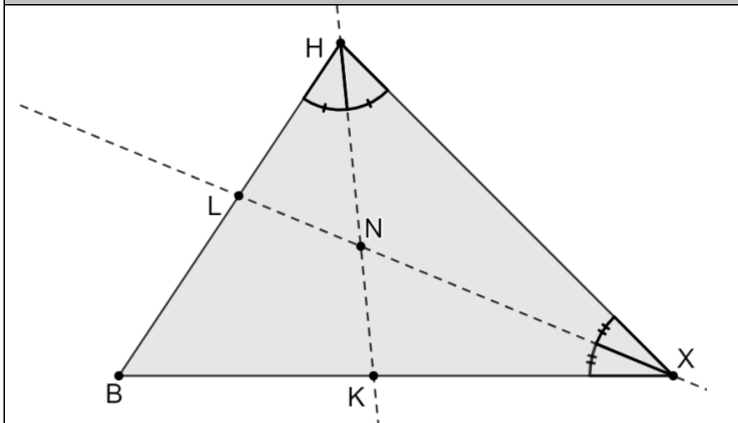
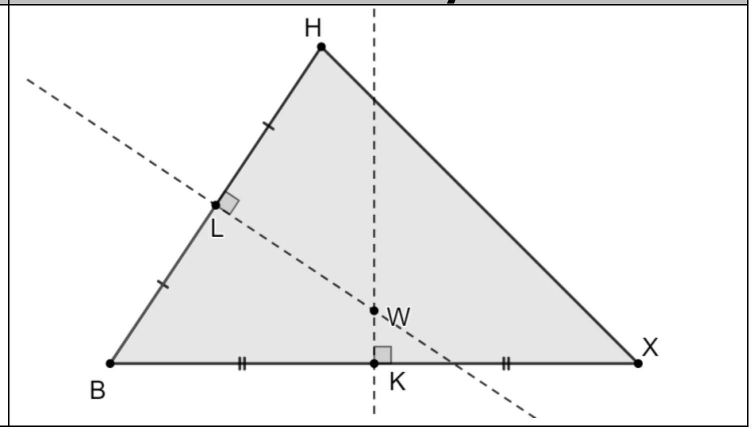
a. Determine the measure of $\overline{RC}$.

$RC =$	

b. Determine the perimeter of ΔYRW.

Perimeter	

4. Nate and Whitney used geometry software for two different constructions. Their incomplete constructions are shown.

Nate	Whitney
	

Complete the statements.

- ○ Nate's
- ○ Whitney's

construction uses

- ○ medians
- ○ angle bisectors
- ○ segment bisectors
- ○ perpendicular bisectors

to construct the incenter of ΔHBX. When completed,

- ○ Nate's
- ○ Whitney's

construction will result in a circle that is circumscribed about ΔHBX with

- ○ $\overline{BN}$
- ○ $\overline{BW}$
- ○ $\overline{LN}$
- ○ $\overline{LW}$

as a radius.

- ○ Nate's
- ○ Whitney's

construction can be used to show the Triangle Sum theorem because the angles of ΔHBX are

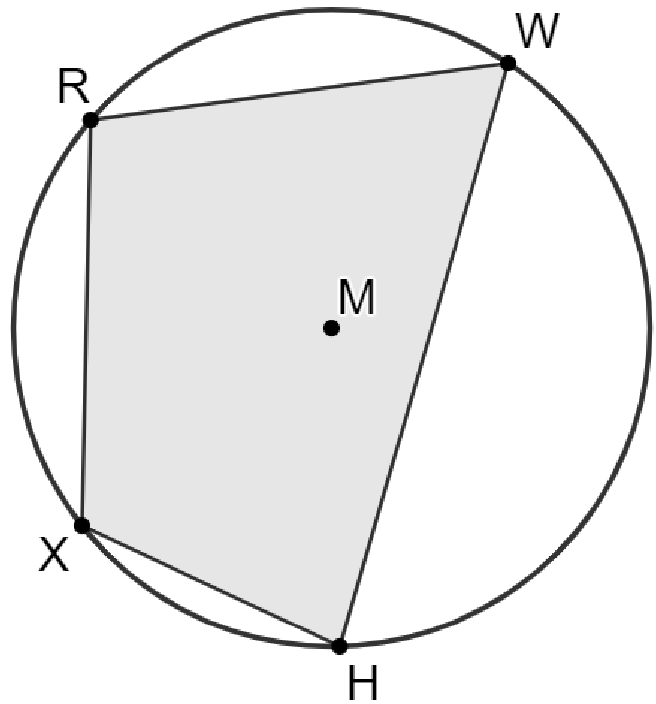
- ○ central angles
- ○ inscribed angles
- ○ angles formed by tangents

of the

- ○ inscribed
- ○ circumscribed

circle such that the three arcs intercepted by the interior angles of ΔHBX complete a circle, so the sum of the measures of the interior angles of ΔHBX is half the measure of a circle.

5. Circle M is shown where points R, W, H, X are on the circle.
$m\angle WRX = (5x + 14)°$,
$m\angle RWH = (3x + 3y)°$,
$m\angle RXH = (2x + 16y)°$, and
$m\angle WHX = (2x + 47)°$.



a. Determine the value of x.

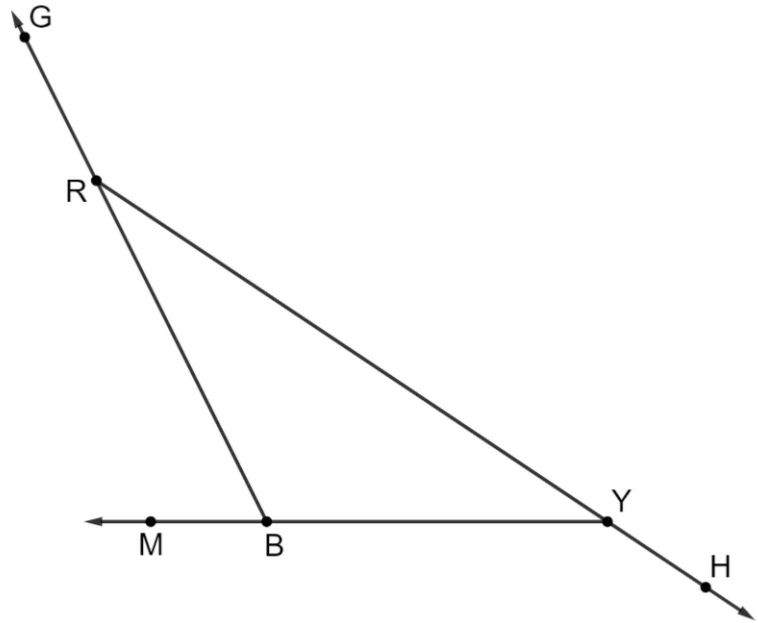
$x =$	

b. Determine the value of y.

$y =$	

6. Triangle RYM is shown where $m\angle HYB = (8d + 9.5)°$, $m\angle BRY = 29.7°$, and $m\angle MBR = (4d - 5)°$.

Determine the value of d.

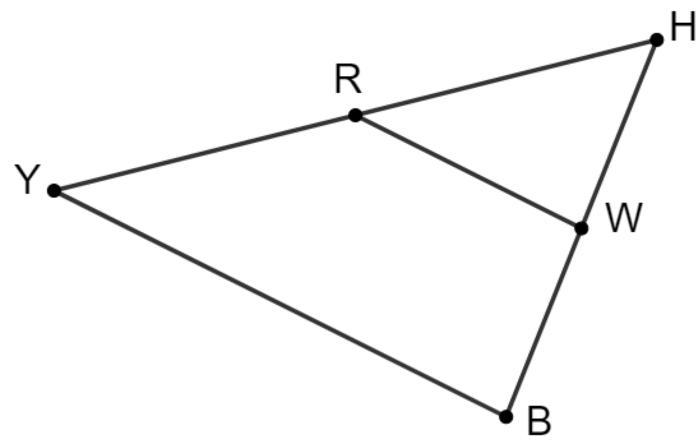


$\boldsymbol{d} =$	

7. Select all of the side lengths that can be used to create a triangle.

- ☐ $14, 14, 14$
- ☐ $17, 18, 35$
- ☐ $\frac{37}{4}, \frac{37}{8}, \frac{37}{12}$
- ☐ $23.1, 45.6, 73.4$
- ☐ $45.62, 65.76, 111.21$

Triangle YHB is shown where $\overline{RW}$ is a midsegment. $YB = 40.25$, $BH = 32.31$, and $YR = 24.74$.



8. Determine the measure of each.

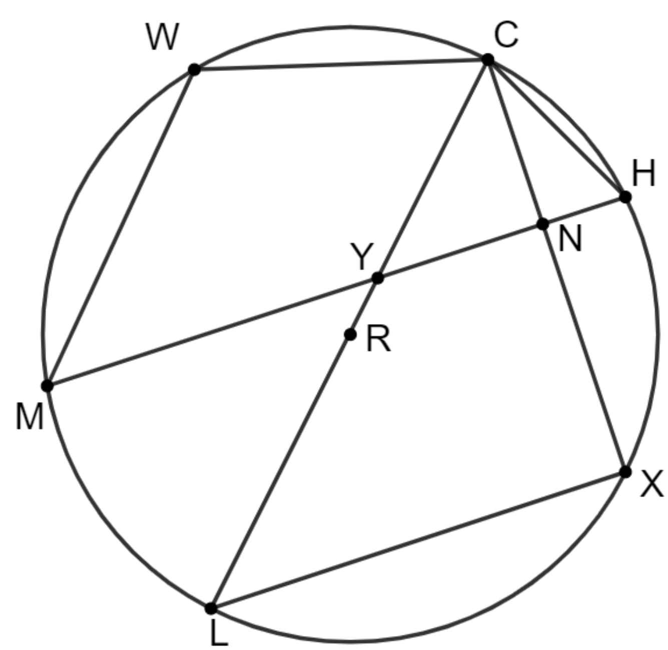
	Segment	Length
a.	$\overline{RW}$	
b.	$\overline{RH}$	

9. Select all of the statements that are true about ΔYHB.

- ☐ $\overline{YH} \cong \overline{HB}$
- ☐ $\overline{YR} \cong \overline{BW}$
- ☐ $\Delta RHW \sim \Delta YHB$
- ☐ $\angle HRW \cong \angle HWR$
- ☐ $\angle HWR \cong \angle WBY$

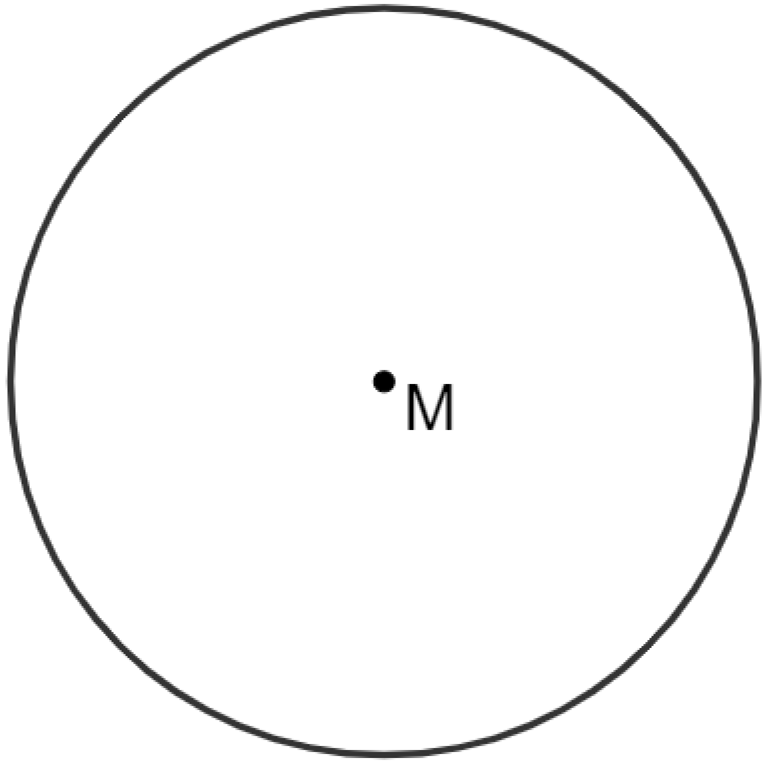
10. Circle R is shown where points C, H, X, L, M, and W are on the circle.

Select all of the statements that must be true.



- ☐ $m\angle LXC = 90°$
- ☐ $m\angle CNH = 90°$
- ☐ $m\angle XNY + m\angle YLX = 180°$
- ☐ $m\angle MWC + m\angle MHC = 180°$
- ☐ $m\angle MWC + m\angle WMH = 180°$

11. Circle M is shown.



Use the circle to construct a hexagon that is inscribed in circle M.